

Experiment No: 1

Aim:

No load and blocked rotor test on a single-phase induction motor (capacitor start type) and determine equivalent circuit parameters.

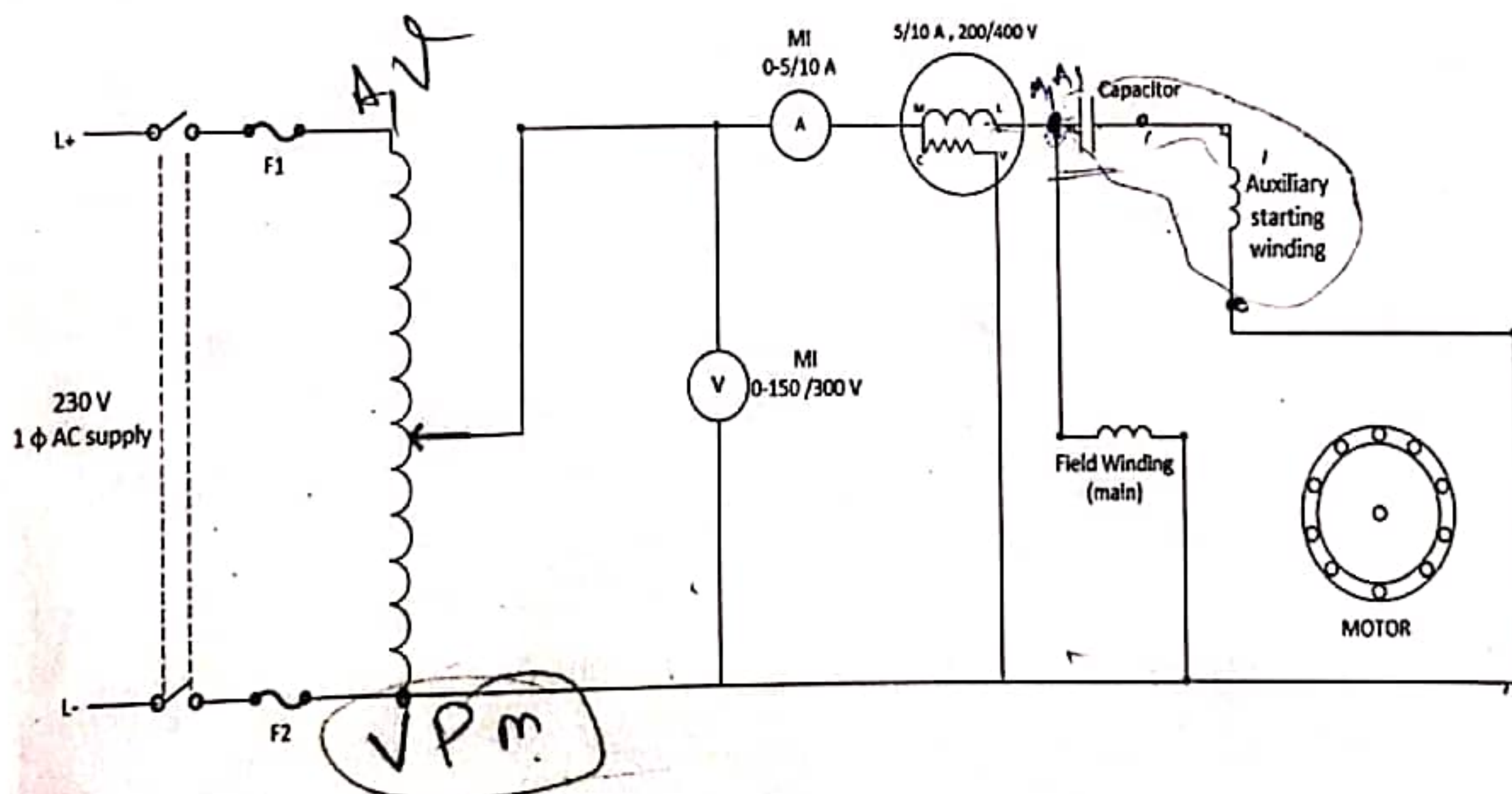
Name Plate Specification of Induction Motor:

230V, 7.4A, 1420 RPM Single phase Induction motor.

Apparatus Required:

S. No.	Apparatus	Specifications	Type	Quantity
1	Auto Transformer			1
2	Ammeter	0-5/10 A	MI	1
3	Voltmeter	0-15/300V	MI	1
4	Wattmeter	5/10 A, 200 /400 V	Dynamometer	1
5	Tachometer			1

Circuit Diagram:



Theory:

No-load and blocked rotor tests are two important tests conducted on a single-phase induction motor to determine its performance characteristics. These tests are usually performed in the laboratory or factory, prior to the motor's installation, to ensure that it meets the desired specifications and standards.

1. **No-load test:** In a no-load test, the motor is operated without any load or mechanical load on its shaft. The purpose of this test is to measure the power input, no-load current, and no-load power factor of the motor. During this test, the motor is connected to a suitable AC supply and allowed to run at its rated speed. The power input to the motor is measured using a wattmeter and the no-load current and power factor are measured using an ammeter and power factor meter, respectively.
2. **Blocked rotor test:** In a blocked rotor test, the motor is operated with its rotor shaft locked or blocked, and with rated voltage applied to the stator winding. The purpose of this test is to measure the motor's input power, current, power factor, and temperature rise at full load. During this test, the motor is connected to a suitable AC supply, and a suitable mechanical load is applied to the motor's shaft to prevent it from rotating. The power input to the motor is measured using a wattmeter, and the current and power factor are measured using an ammeter and power factor meter, respectively. The temperature rise of the motor is measured using a thermometer, and this information is used to determine the motor's efficiency.

Based on the results obtained from these tests, the motor's performance characteristics such as its efficiency, power factor, and starting torque can be determined. These tests also help in identifying any defects or issues in the motor and ensure that it is functioning optimally before it is put to use.

The Equivalent circuit of the single-phase induction motor on the basis of double revolving field during theory is shown in figure 2.

To determine the parameters of this equivalent circuit the following tests are necessary.

1. **Measurement of dc resistance r_{10} of the main winding at the room temperature t_0 .** This is done by voltmeter ammeter method.
Its resistance at the operating temperature is given by-
$$r_1 = r_{10} [1 + \alpha(t - t_0)]$$
2. **A blocked rotor test is performed at rated current and frequency (with the starting winding open).** The reading of V_b , I_b and W_b are noted.
For this condition as $s=1$, the equivalent circuit simplifies to figure 3.
3. **In no load test, the motor is run at rated voltage and frequency.** The slip is extremely small under no load operations and thus the equivalent circuit under these conditions simplifies to figure 4. V_0 , I_0 , and W_0 are noted down with this test.

Procedure:

NO LOAD TEST -

1. Connect the circuit as shown in figure 1.
2. Switch ON the AC supply to the circuit.
3. Adjust the voltage and then increase it to the t-rated voltage of the motor.
4. Record the applied voltage V_0 , no load current I_0 and no load power W_0 .
5. Reduce the applied voltage to zero and switch OFF the supply to stop the motor.

BLOCKED ROTOR TEST -

1. Connect the circuit as above and disconnect starting winding from circuit and switch ON the supply.
2. Adjust the variac such that rated current flows in the motor.
3. Record applied voltage V_{sc} , rated input current I_{sc} and Power W_{sc} .
4. Switch OFF the AC mains to stop the motor.

RESISTANCE OF MAIN WINDING -

Measure the dc resistance of the main stator winding using voltmeter ammeter method.

Observations:

NO LOAD TEST			BLOCKED ROTOR TEST		
V_0	I_0	W_0	V_{sc}	I_{sc}	W_{sc}

Calculations:

$$Z_{sc} = V_{sc}/I_{sc}$$

$$R_{sc} = W_{sc}/I_{sc}$$

$$X_{sc} = (Z_{sc}^2 - R_{sc}^2)^{1/2}$$

$$Z_0 = V_0/I_0$$

$$\cos\phi_0 = W_0/V_0I_0$$

Result:

Precautions:

1. Slowly increase the voltage and note the readings carefully.
2. Connect the ammeter and voltmeter with correct polarity.